

BDX Slow Control Alarm Reference

Operational meaning, thresholds, delays, notification routing, and transient-suppression rules
Configuration reference: **devel** branch, 1 October 2026

Purpose. This document summarizes the alarm rules implemented by the BDX slow-control notifier. Repeated per-channel and per-sensor rules are consolidated into rule families; their scope is stated explicitly. The authoritative machine-readable configuration remains `notifier/alarms.json`.

1. Severity and notification policy

Severity	Meaning	Notification
MINOR	Abnormal condition requiring operator attention, but not an immediate safety action.	Telegram
MAJOR	Persistent or more severe abnormal condition requiring prompt expert attention.	Telegram + email
INTERLOCK	Safety condition associated with an automatic or explicit protective action.	Telegram + email

Unless a rule overrides it, the default numeric policy is: MINOR at 5% for 5 s; direct MAJOR above 10% immediately; sustained MAJOR at 5% for 20 s; recovery after 5 s back in range. Alarm groups suppress duplicate notifications when several rules describe the same underlying condition.

2. Global system alarms

Alarm / PV	Trigger	Severity / delay	Operational meaning
Global interlock BDX:GLOBAL:INTERLOCK_ACTIVE	Interlock becomes active.	INTERLOCK immediately; recovery 1 s	The global interlock has switched the system to its protective state.
Global not ready BDX:GLOBAL:READY	READY = 0 while system state is neither STANDBY nor SAFE.	MINOR after 5 s; MAJOR after 120 s	The system is expected to be operational but one or more readiness conditions are not satisfied.
Global heartbeat stale BDX:GLOBAL:HEARTBEAT	Heartbeat unchanged for 10 s.	MINOR at 10 s stale; MAJOR 50 s later	The global IOC may be frozen, disconnected, or no longer executing its update loop.

3. Chiller alarms

Temperature-deviation monitoring is enabled only while the chiller is running and DEVIATION_SETTLING is false. After a setpoint change or RUN->ON, settling remains active until $|T - Tset|$ is below **0.3 degC for 10 consecutive seconds**, with a maximum settling watchdog of **5 minutes**.

Alarm / PV	Trigger	Severity / delay	Meaning
Temperature deviation TEMPERATURE_RBV vs SETPOINT_RBV	Relative deviation from applied setpoint after settling.	Default numeric policy: MINOR 5%/5 s; MAJOR >10% immediate or 5% sustained 20 s	Bath/control temperature is no longer tracking the requested operating temperature.
Communication unavailable COMM_OK	COMM_OK = 0	MINOR 10 s; MAJOR 60 s	TCP/device communication with the chiller is unavailable.
Heartbeat stale HEARTBEAT	No heartbeat change for 10 s	MINOR then MAJOR after 50 s additional	IOC update loop appears stalled.
IOC not running IOC_STATE	IOC_STATE != RUNNING	MAJOR 5 s	The chiller IOC process is not in its normal RUNNING state.
IOC error code ERROR_CODE	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s	Software/driver layer reports an error.
Coolant level low STAT_LOW_LEVEL	Hardware low-level bit active	MAJOR 60 s; INTERLOCK 600 s	Insufficient coolant level. Persistent low level is treated as an interlock condition.
Overtemperature STAT_OVERTEMPERATURE	Hardware overtemperature bit active	MAJOR 60 s	The chiller reports an internal overtemperature condition.
General STAT bits GENERAL_ERROR / GENERAL_ALARM / GENERAL_WARNING / RESERVED / EXTERNAL_CONTROL_MISSING	Corresponding STAT bit active	MINOR 10 s	Device-reported warning/error/status anomaly requiring inspection.

4. Low-voltage power-supply alarms

Scope: **LV1 CH1, LV1 CH2, LV2 CH1, LV2 CH2**. The CPX400DP meter resolution is 10 mV and 10 mA; practical minimum significant changes are therefore set to **30 mV** and **30 mA**. This avoids large relative percentages caused by quantization near zero. The 30 mA and 30 mV cutoffs are intentionally conservative compared with the instrument meter resolution and fixed accuracy terms.

Rule family	Trigger	Severity / delay	Meaning / suppression
Voltage deviation VOLTAGE_RBV vs VOLTAGE_SET_RBV	$ \Delta V \geq 30 \text{ mV}$ and relative deviation $\geq 5\%$ while output is ON. A 0.2 V minimum reference is used near zero.	MINOR after 10 s; MAJOR after 60 s sustained	Measured voltage is persistently different from the applied setpoint. Small differences below 30 mV are ignored.
Output mismatch OUTPUT_RBV vs OUTPUT_SET	Readback differs from commanded output state.	MAJOR after 5 s	Output state does not match the command. Monitoring is disarmed when OUTPUT_SET changes and rearmed only after the next hardware poll (OUTPUT_MONITOR_READY).
OVP proximity VOLTAGE_RBV / OVP_RBV	Voltage $\geq 95\%$ of OVP; OVP reference must be $\geq 30 \text{ mV}$.	MINOR after 5 s at $\geq 95\%$; MAJOR immediately at $\geq 100\%$	Operating voltage is approaching or exceeding the configured over-voltage protection threshold.
Hardware OCP trip OCP_TRIPPED	Hardware OCP trip bit active.	MINOR immediately; MAJOR 10 s; INTERLOCK 20 s	The PSU hardware has tripped on over-current. At INTERLOCK stage the slow control explicitly commands the output OFF.
Fast current change CURRENT_CHANGE_2S_PERCENT	Current changes $\geq 10\%$ within 2 s, but only if $ \Delta I \geq 30 \text{ mA}$.	MAJOR immediately	Abrupt current excursion, potentially indicating a load change, short, disconnect, or instability.
Slow current change CURRENT_CHANGE_10S_PERCENT	Current changes $\geq 10\%$ within 10 s, with $ \Delta I \geq 30 \text{ mA}$.	MINOR immediately	Slower but significant current drift.

Rule family	Trigger	Severity / delay	Meaning / suppression
Voltage span - minor VOLTAGE_SPAN_30S	30 s readback span >0.1 V.	MINOR immediately once 30 s window is available	Voltage is not stable over the rolling 30 s window.
Voltage span - major VOLTAGE_SPAN_30S	30 s readback span >0.5 V.	MAJOR immediately once 30 s window is available	Large voltage instability over the rolling 30 s window.

Transient handling for LV diagnostics

Current-change and rolling-span diagnostics clear their history when output state, voltage setpoint, or current limit changes. They then observe a 5 s diagnostic grace period before rebuilding the rolling baseline. When the output is OFF, these dynamic metrics are published as unavailable (NaN) and do not generate alarms.

Device-level LV health

Alarm	Trigger	Severity / delay	Meaning
LV1/LV2 communication	COMM_OK = 0	MINOR 10 s; MAJOR 60 s	Power-supply communication unavailable.
LV1/LV2 heartbeat stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 50 s additional	IOC update loop appears stalled.
LV1/LV2 IOC state	IOC_STATE != RUNNING	MAJOR 5 s	IOC is not running normally.
LV1/LV2 error code	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s	Driver/IOC reports an error.

5. High-voltage alarms (optional subsystem)

HV1 is optional in the current prototype configuration. The generic channel template enables voltage deviation, output mismatch, and OVP proximity for HV1 CH1/CH2. LV-specific OCP-trip, current-change, and 30 s voltage-span diagnostics are disabled for HV channels.

Alarm	Trigger	Severity / delay	Meaning
HV1 communication	COMM_OK = 0	MAJOR 5 s	HV communication unavailable.
HV1 heartbeat stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 10 s additional	HV IOC appears stalled.
HV1 IOC state	IOC_STATE != RUNNING	MAJOR 5 s	HV IOC not running normally.
HV1 error code	ERROR_CODE != 0	MINOR 10 s; MAJOR 60 s	HV IOC/driver reports an error.
HV1 CH1/CH2 voltage deviation	Same generic voltage-deviation rule; output must be ON.	MINOR 10 s; MAJOR 60 s sustained	Measured HV differs from commanded setpoint.
HV1 CH1/CH2 output mismatch	OUTPUT_RBV != OUTPUT_SET	MAJOR 5 s	Hardware output state differs from command.
HV1 CH1/CH2 OVP proximity	Voltage >=95% or 100% of OVP; OVP >=30 mV	MINOR 5 s / MAJOR immediate	Voltage approaches or reaches OVP threshold.

6. Environmental temperature alarms

Scope: temperature sensors **T00**, **T01**, **T02**, **T03**. The same rule family is instantiated for each sensor.

Rule family	Trigger	Severity / delay	Meaning
Comfort range	T <15 degC or T >28 degC	MINOR immediately	Temperature is outside the preferred monitoring band but not yet at a critical absolute limit.
Critical range	T <10 degC or T >35 degC	MAJOR after 5 min	Temperature has remained outside the critical absolute range.
10 min temperature change	CHANGE_10M >=3 degC	MAJOR immediately	Rapid thermal drift over 10 minutes.
1 h temperature change	CHANGE_1H >=5 degC	MAJOR immediately	Large cumulative thermal drift over one hour.
Sensor unavailable	STATUS_OK = 0	MINOR 10 s; MAJOR 60 s	Sensor or communication path is unavailable.
Sensor stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 50 s additional	Sensor process is connected but no longer updating.
Inter-sensor spread	TEMPERATURE_SPREAD >4 degC	MINOR after 1 min; MAJOR after 20 min	Temperature sensors disagree significantly, suggesting a gradient, local heating, or sensor fault.

7. Other environmental sensors

Scope / alarm	Trigger	Severity / delay	Meaning
Humidity H01 - health	STATUS_OK = 0	MINOR 5 s; MAJOR 20 s	Humidity sensor unavailable.
Humidity H01 - stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 15 s additional	Humidity sensor stopped updating.
Pressure P01 - health	STATUS_OK = 0	MINOR 5 s; MAJOR 20 s	Pressure sensor unavailable.
Pressure P01 - stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 15 s additional	Pressure sensor stopped updating.

8. Archiver alarms

Alarm	Trigger	Severity / delay	Meaning
Overall health BDX:ARCHIVER:OK	OK = 0	MINOR 10 s; MAJOR 60 s	Archiver Appliance is unavailable or degraded.
Health-check stale LAST_CHECK	No change for 30 s	MINOR at 30 s stale; MAJOR 30 s later	Archiver health monitor itself has stopped updating.
Required PV coverage	REQUIRED_HEALTHY < REQUIRED_TOTAL	MINOR 10 s; MAJOR 60 s	At least one PV required by the deployment is not healthy in the Archiver.

9. DAQ alarms (optional subsystem)

Alarm	Trigger	Severity / delay	Meaning
DAQ communication	COMM_OK = 0	MAJOR 5 s	DAQ crate communication unavailable.
DAQ heartbeat stale	Heartbeat unchanged 10 s	MINOR then MAJOR after 10 s additional	DAQ IOC appears stalled.
DAQ IOC state	IOC_STATE != RUNNING	MAJOR 5 s	DAQ IOC not running normally.
DAQ error	ERROR non-empty	MINOR 5 s; MAJOR 20 s	DAQ IOC reports an explicit error string.
DAQ state mismatch	STATE_RBV != EXPECTED_STATE_RBV	MINOR 5 s; MAJOR 20 s	Actual crate state differs from the state requested/expected by the control system.

Alarm	Trigger	Severity / delay	Meaning
DAQ not ready during measurement	READY = 0 while MEASUREMENT_ACTIVE = 1	MAJOR immediately	DAQ lost readiness during an active measurement.

10. Interpretation notes

Percentages near zero

Relative differences are meaningful only when the reference and the absolute excursion are above the measurement floor. Current dynamics require $|\Delta I| \geq 30$ mA. Voltage deviation requires $|\Delta V| \geq 30$ mV and uses a 0.2 V minimum denominator near zero. OVP proximity is inactive for OVP references below 30 mV.

Commanded transitions

Expected transients caused by operator commands are explicitly suppressed where appropriate. LV dynamic diagnostics have a 5 s grace period after control changes; output mismatch is armed only after the first hardware poll following an output command; chiller deviation monitoring uses condition-based settling.

Recovery

Most alarms require the condition to remain healthy for a configured recovery interval before a resolution notification is emitted. Typical recovery is 5 s; global interlock uses 1 s, archiver rules 10 s, and the global heartbeat 2 s.

Optional alarms

Rules marked optional do not prevent notifier startup if their PVs are absent. This is used for subsystems not deployed in the current prototype, notably HV, DAQ, humidity, pressure, and some environmental sensors.

Source of truth

This reference is descriptive. The executable thresholds and rule wiring are defined in `notifier/alarms.json` and the IOC diagnostic logic in `src/bdx_slow_control/iocs/`.

11. Instrument basis for LV measurement floors

The Aim-TTi CPX400D/DP manual specifies dual 4-digit meters with a 4 Hz reading rate, 10 mV / 10 mA meter resolution, voltage accuracy of 0.1% of reading ± 2 digits, and current accuracy of 0.3% of reading ± 2 digits. The 30 mV and 30 mA minimum-significant-change thresholds are therefore deliberately set above a single display/readback count and the fixed two-digit accuracy term.

Reference: Aim-TTi, *CPX400D & DP Instruction Manual*, current product/support documentation.
<https://www.aimtti.com/product-category/dc-power-supplies/aim-cpxseries>

Document maintenance

Update this document whenever alarm thresholds, severity routing, settling/grace behavior, or the set of deployed subsystems changes. A recommended review point is each stable release promoted from devel to main.